Product Specification
产品规格书

Part NO./产品型号：**XY_H2EDP_UHD95T_V3.1**

Product Code/产品编码：**A00401**

Update time/更新时间：**2022.04.01**

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1.General Description / 概述.

XY_H2EDP_UHD95_V3.1 is a multi-function display driver board combines 2 HDMI2.0 and 1 DP1.2 input interface, supports 4-lane eDP HBR2 output up to 4096x2160@60Hz, supports headphone output and 3 watts dual channel audio amplifier output.

XY_H2EDP_UHD95_V3.1 是一款多功能的显示器驱动板，2 HDMI2.0 和 1 DP1.2 输入，4 路 eDP（HBR2）输出，最高支持 4096x2160@60Hz，支持耳机输出和 3 瓦特双声道音频功放输出。

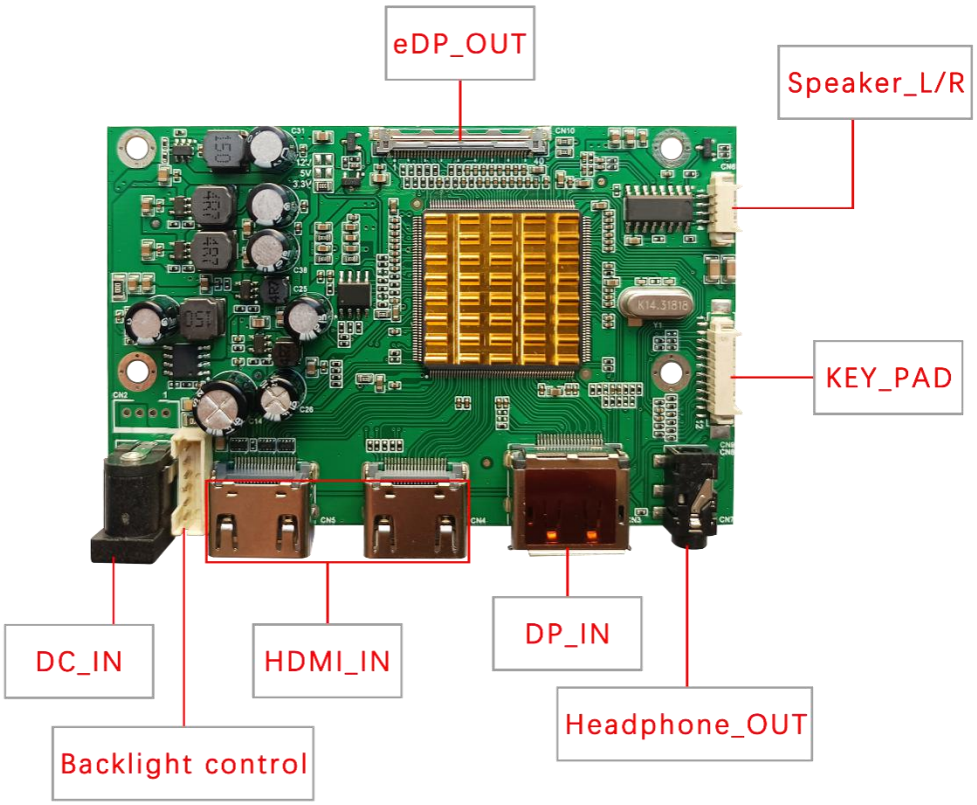
2.Features / 特性.

In the following table you will find the detailed features/如下表.

Power input/电源输入	12V 电源输入
Signal Input/信号输入	2 HDMI Input / 2 路 HDMI 输入
Signal Input/信号输入	1 DP Input / 1 路 DP 输入
Signal Output/信号输出	1 eDP Input / 1 路 eDP 输出
Audio output/音频输出	3.5mm headphone & 3W speaker/耳机和功放
Keypad/按键	Power/Menu/Right/Left/Exit, 5 key
Built-in LED Driver/背光驱动	up to 20W/最高 20W
……	……

3.Interface Function Description Diagram / 接口功能描述.

NO./编号	Description/描述
CN1	Power input / 12V 电源输入
CN2	Built_in Power_in / 内置电源输入
CN3	DP input / DP 输入
CN4/CN5	HDMI input / HDMI 输入
CN6	Speaker_L/R / 功放输出
CN7	Headphone output / 3.5mm 耳机输出
CN9	Keypad interface / 按键接口
CN10	eDP output / eDP 输出
CN11	Backlight control / 背光控制接口

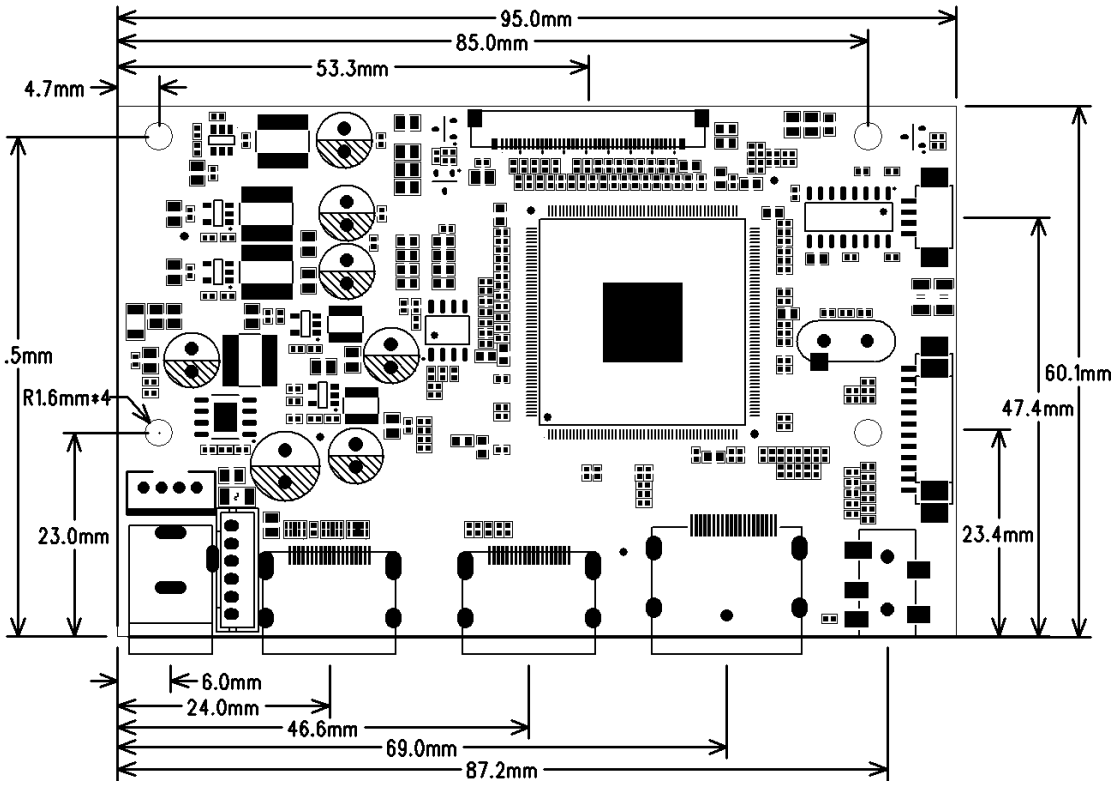


4.Structural Dimension Diagram / 结构尺寸.

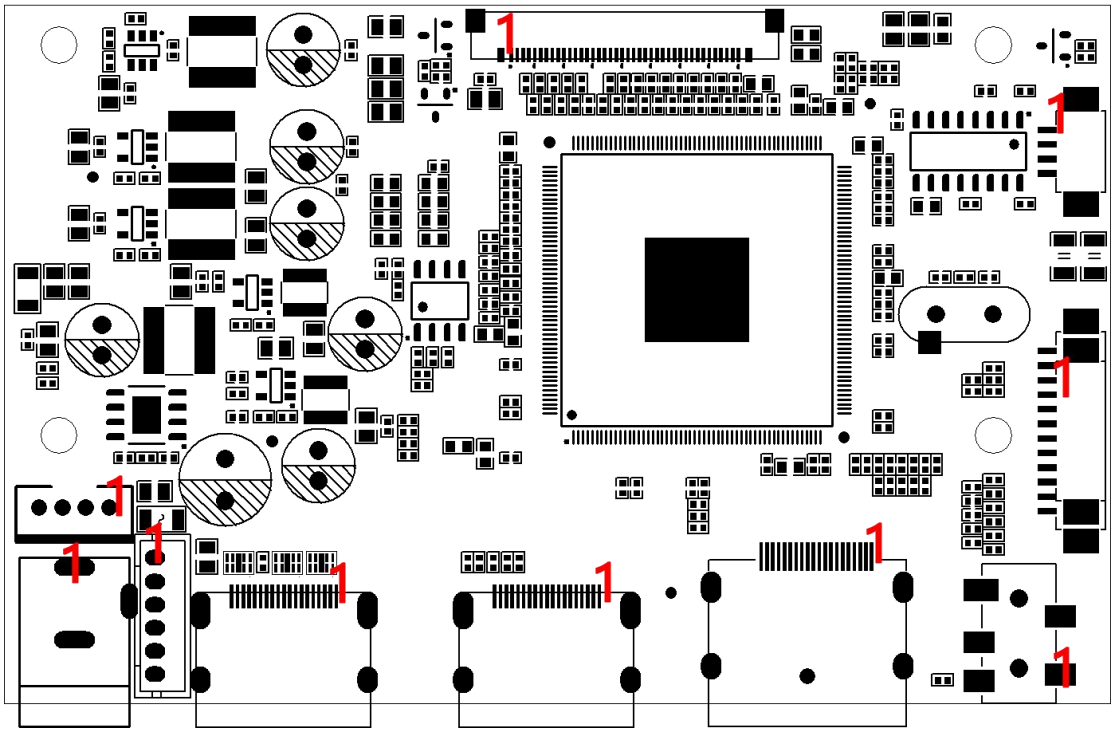
Unit/单位：mm

Height/高度: 12.6mm

Tolerance/公差：±0.3mm



5.Interface Definition / 接口定义.



1.CN1：Power input / 12V 电源输入

Pin NO.	Pin Name	Description
1	12V	12V Power Input
2	GND	Ground
3	GND	Ground

2.CN2：Power input / 12V 内置电源输入

P/N	Symbol	Function
1	GND	Ground
2	GND	Ground
3	12V	12V Power Input
4	12V	12V Power Input

4.CN4/CN5：HDMI input / HDMI 输入

P/N	Symbol	Function
1	DATA2+	HDMI Differential Signal Data Input 2+
2	DATA2 SHIELD	HDMI Differential Signal Data Input 2 Shield

3	DATA2-	HDMI Differential Signal Data Input 2-
4	DATA1+	HDMI Differential Signal Data Input 1+
5	DATA1 SHIELD	HDMI Differential Signal Data Input 1 Shield
6	DATA1-	HDMI Differential Signal Data Input 1-
7	DATA0+	HDMI Differential Signal Data Input 0+
8	DATA0 SHIELD	HDMI Differential Signal Data Input 0 Shield
9	DATA0-	HDMI Differential Signal Data Input 0-
10	CLK+	HDMI Differential Signal Clock+
11	CLK SHIELD	HDMI Differential Signal Clock Shield
12	CLK-	HDMI Differential Signal Clock-
13	CEC	CEC Control
14	NC	NO Connect
15	SCL	DDC Clock Line
16	SDA	DDC Data Line
17	GND	Ground
18	+5V POWER	+5V Power Line
19	HOT PLUG	Hot Plug Detect

5.CN6：Speaker_L/R / 功放输出

Pin NO.	Pin Name	Description
1	L+	Left Vocal+
2	L-	Left Vocal-
3	R-	Right Vocal-
4	R+	Right Vocal+

6.CN7：Headphone output / 3.5mm 耳机输出

Pin NO.	Pin Name	Description
1	GND	Ground
2	GND	Ground
3	AUDIO_HOUTR	AUDIO_HOUTR
4	AUDIO_HDET	AUDIO_HDET
5	AUDIO_HOUTL	AUDIO_HOUTL

7.CN9：Keypad interface / 按键接口

Pin NO.	Pin Name	Description
1	POWER	Power Key
2	LEDR	LED Red
3	LEDG	LED Green
4	GND	Ground

5	LEFT	LEFT Key
6	RIGHT	RIGHT Key
7	AUTO	Signal Source Switching
8	MENU	Menu Key

8. CN10：eDP output / eDP 输出

Pin NO.	Pin Name	Description
1	COLOR_E	COLOR_E
2	VCC_LED	LED Power
3	VCC_LED	LED Power
4	VCC_LED	LED Power
5	VCC_LED	LED Power
6	NC	NO Connect
7	NC	NO Connect
8	BL_PWM	LED Backlight Adjust
9	BL_EN	LED Backlight Enable
10	GND	Ground
11	GND	Ground
12	GND	Ground
13	GND	Ground
14	HPD	Hot Plug Detect
15	GND	Ground
16	GND	Ground
17	GND	Ground
18	GND	Ground
19	LCD_BIST	LCD_BIST
20	PANEL_VCC	Panel Power
21	PANEL_VCC	Panel Power
22	PANEL_VCC	Panel Power
23	PANEL_VCC	Panel Power
24	GND	Ground
25	AUXN	Component Signal for Auxiliary Channel
26	AUXP	True Signal for Auxiliary Channel
27	GND	Ground
28	TX0P	Display Port Lane 0+ Output
29	TX0N	Display Port Lane 0- Output
30	GND	Ground
31	TX1P	Display Port Lane 1+ Output
32	TX1N	Display Port Lane 1- Output
33	GND	Ground

34	TX2P	Display Port Lane 2+ Output
35	TX2N	Display Port Lane 2- Output
36	GND	Ground
37	TX3P	Display Port Lane 3+ Output
38	TX3N	Display Port Lane 3- Output
39	GND	Ground
40	NC	NO Connect

9. CN11：Backlight control / 背光控制接口

Pin NO.	Pin Name	Description
1	12V	12V DC Power Supply
2	12V	12V DC Power Supply
3	EN	Backlight Enable
4	PWM	Backlight Brightness Adjust
5	GND	Ground
6	GND	Ground

6.Operation Requirements / 使用环境.

- Relative humidity/湿度: ≤ 80%
- Storage temperature/存储温度: 0 ~ +40℃
- Operation temperature/工作温度: 0 ~ +40℃

7.General Precautions / 注意事项.

- Keep the board away from strong electrostatic.保持电路板远离强静电。
- Disconnect main power if power of panel is incorrect, otherwise you would destroyed the LCD module.如果面板电源不正确，请断开主电源，否则会损坏 LCD 模块。
- Avoid dropping metal in the board while it is working.在电路板工作时，避免金属掉落在电路板中。
- Do not push or pull the connector while the board is working.请勿在电路板工作时推拉连接器。
- Do not disassemble the module.请勿拆解模块。
- If the surface of the main board is dirty, clean it with soft (not wet) cloth. 如果主板表面很脏，请用软布（非湿布）清洁。

8.Revision History / 版本更新

Rev./版本号	Date/日期	Note/备注
V1.0	2019-08-19	
V3.0	2022-03-07	
V3.1	2022-03-23	
……	……	